6.4 Fermentation

Fermentation is one of the age old method of food preservation techniques. Fermentation extends the shelf life of the foods similar to that of salting and drying. The term fermentation refers to both the anaerobic and aerobic breakdowns of carbohydrates and carbohydrate-like materials. Fermentation means the process of the reaction between microorganisms and organic compounds in natural conditions. Apart from carbohydrate, microorganisms and enzymes react on protein and fat by releasing carbon-di-oxide and other gases.

a. Anaerobic condition: In cheese production, due to Streptococcus lactis bacteria, lactose is converted to lactic acid under anaerobic condition.

b. Aerobic condition: In vinegar production, due to Acetobacter bacteria ethyl alcohol is oxidised to acetic acid in aerobic condition.

Foods Produced by Fermentation

1. Alcoholic Beverages

Wine: Yeast is present on the skin of grapes. Fermentation process starts when yeast reacts with the sugars of grapes to convert them into alcohol.

Beer and Ale: These contain malted cereals which are fermented by yeast to yield 3–7% of alcohol. The type of yeast, quantity and fermentation temperature control the alcohol production.

2. Vinegar Preparation: In the presence of oxygen, *Acetobacter* bacteria convert alcohol to acetic acid. Vinegar can be prepared from carbohydrate sources like cereals and fruits.

3. Cheese Production: *Streptococcus lactis* bacteria changes milk sugar into lactic acid. This acid curdles milk into cheese.

4. Citric acid Production: Citric acid is produced from yeast, mould and bacteria. It is used in the production of fruit beverages.

Uses of Fermentation

1. Fermentation extends the shelf life of foods.
2. It enhances the growth of microorganisms producing acid and alcohol.
3. It prevents the growth of lipolytic and proteolytic microorganisms.
4. Vinegar obtained by fermentation has industrial importance.

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